Dinithi Jayasuriya

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Professional Summary

PhD Research Assistant in Electrical and Computer Engineering focused on AI-guided electronic design automation for digital physical design and analog circuit evaluation. Previously worked at Synopsys on ASIC and SoC verification in Clock Domain Crossing. Strengths include hardware validation and debugging, RTL fundamentals in SystemVerilog and Verilog, SPICE-based circuit analysis, and building scripting-driven evaluation pipelines for power, performance, area, and overall quality of results. Current work studies reliability and uncertainty in LLM-assisted OpenROAD workflows using controlled experiments, failure analysis, and metric-driven optimization. Published researcher with 21 papers and 60 citations as of April 2026, aiming to contribute to silicon hardware engineering, verification and validation, and semiconductor research.

Education

PhD **University of Illinois, Chicago**, Electrical and Computer Engineering — Anticipated in 2028

- CGPA: 3.9/4.0
- **Relevant Coursework:** VLSI Designs, Digital System Designs, Advanced Microprocessor Architecture, Causal Inference, HDL

BSc **University of Sri Jayewardenepura, Sri Lanka**, Electrical and Electronics Engineering — Nov 2016 – Aug 2021

- GPA: 3.64/4.0

Technical Skills

Silicon / Hardware: RTL fundamentals, SystemVerilog, Verilog, assertions (SVA), CDC concepts, ASIC/SoC verification workflows, debug/root-cause analysis

Physical Design / PPA: OpenROAD physical design flow (PnR), QoR/PPA reporting (timing/constraint awareness, congestion/overflow, wirelength trends), experiment-driven optimization

Analog / Circuits: SPICE simulations (DC/AC/Transient/Noise), corner-style evaluation mindset, metric extraction (gain, UGBW, phase margin, power, noise, slew rate)

Programming / Scripting: Python (automation, parsing, analysis, experiment orchestration), Bash, Git, Linux; familiarity with EDA-style scripting patterns used in Tcl-based flows

Research / Methods: Design of Experiments (DoE) comparisons, failure analysis, data analysis, reproducibility, technical documentation, code review habits

AI/ML (applied to EDA): LLM-assisted tool scripting, agentic diagnose–fix–rerun loops, causal inference / SCMs for attribution and decision support

Research Experience

University of Illinois Chicago, Ph.D. Research Assistant (EDA + AI) — Chicago, IL, Aug 2023 – Present

- Initiate and execute research projects at the intersection of EDA and AI: define hypotheses, design controlled evaluations, and communicate results via papers and presentations.
- Develop tooling to analyze intermediate EDA signals (reports/logs) for debugging, failure mode classification, and metric-driven optimization across design stages.

Professional Experience

Synopsys Inc, Application Engineer (Clock Domain Crossing / Verification) — Sri Lanka, Aug 2021 – Jul 2023

- Supported advanced ASIC/SoC verification flows focused on CDC correctness; debugged complex customer-reported issues and delivered clear root-cause analysis with actionable fixes/workarounds.

- Owned end-to-end customer enablement flows for Apple and Renesas, emphasizing reproducible testcases, documentation, and quality-driven escalation handling with R&D teams.
- Built and maintained a regression-style validation workflow for CDC issues: reproduce customer failures with minimal testcases, define pass/fail checks, run controlled experiments across tool settings and design variants, and document results to confirm fixes before customer delivery.
- Validated a glitch-identification feature for Apple using stress testing and systematic test design; achieved 98% accuracy and received an internal performance scholarship for high-quality delivery.
- Strengthened engineering rigor through code-review mindset, detailed technical documentation, and structured debug notes to reduce re-escalations.

Mobitel (Pvt) Ltd, Electronic Engineer Intern — Sri Lanka, Mar 2020 – Sep 2020

- Exposure to production troubleshooting and operational validation in telecommunications infrastructure; practiced systematic debugging and documentation in a live environment.

Projects

Verifier-Guided EDA Code Generation in OpenROAD (ICCAD 2026 Submission – Under Review) — Jan 2026 – Present

- Developed a verifier-guided framework for LLM-based OpenROAD code generation that enforces structural correctness before execution, reducing reliance on tool-in-the-loop debugging.
- Modeled OpenROAD query–action tasks as structural dependency graphs over design objects and used them to guide retrieval, constrained code generation, and staged pre-execution verification.
- Built a multi-layer verification and diagnosis-driven repair pipeline to detect invalid acquisition paths, missing prerequisites, API mismatches, and semantic inconsistencies in generated scripts.
- Evaluated the framework on single-step and multi-step OpenROAD tasks; improved single-step pass rate to 82.5% and multi-step pass rate to 84.0% with trajectory-level reflection, while reducing tool calls by more than $2\times$.
- Designed post-verification uncertainty estimation to filter unreliable candidates, improving verifier precision from 80.0% to 93.3% and reducing false positives from 20.0% to 6.7%.

Causal Structure Learning for Scalable Analog Circuit Optimization (SPICE) — Aug 2024 – Present

- Researching how to infer the causal structure of analog circuits by relating design parameters and internal operating-point behaviors to final performance metrics, with the goal of enabling more interpretable and sample-efficient optimization.
- Developing causal models that capture which parameters and intermediate electrical conditions drive key trade-offs, so optimization can generalize to larger circuits without relying heavily on manual expert tuning.
- Studying technology-transfer challenges by analyzing how causal relationships shift across different operating conditions and variation settings, and using the learned structure to adapt optimization strategies with reduced human effort.
- Running SPICE-based evaluation loops to validate hypotheses and quantify improvements using controlled comparisons, tracking multi-objective targets such as gain, bandwidth, stability margins, power, and noise.

Dynamic Rank Learning for Quantized Foundation Models — Aug 2024

- Enhanced Low-Rank Adaptation (LoRA) by enabling dynamic rank learning; emphasized accuracy–efficiency tradeoffs relevant to performance-aware compute systems.

Gesture Controlled Smart Room Kit May 2020

- Developed an undergraduate embedded systems project; placed 7th in Robotics and Embedded Systems category at the Innovation and Invention Exhibition (University of Sri Jayewardenepura).

Publications

Structural Verification for Reliable EDA Code Generation without Tool-in-the-Loop Debugging Under Review
Submitted to ICCAD 2026; arXiv:2604.18834
arXiv link

CaDRO: Causal-Guided Dimensionality Reduction for Scalable Multi-Objective Pareto Optimization Nov 2025
Design, Automation & Test in Europe (DATE)
arXiv link

Gaussian-Sigmoid Reinforcement Transistors: Resolving Exploration-Exploitation Trade-Off Through Gate Voltage-Controlled Activation Functions Jul 2025
Advanced Functional Materials
Publisher link

Sparc: Subspace-aware Prompt Adaptation for Robust Continual Learning in LLMs Jun 2025
IJCNN 2025
arXiv link

Enhancing 3D Robotic Vision Robustness by Minimizing Adversarial Mutual Information through Curriculum Training May 2025
ICRA 2025
arXiv link

Neural Precision Polarization: Simplifying Neural Network Inference with Dual-Level Precision Nov 2024
IEEE ISCAS 2025
ResearchGate link

Single-Step Extraction of Transformer Attention With Dual-Gated Memtransistor Crossbars Oct 2024
IEEE Electron Device Letters
ResearchGate link

Leadership Experience

Head of the Women's Forum Synopsys Inc, Sri Lanka
President: Electrical and Electronic Club University of Sri Jayewardenepura

- Led the Electrical and Electronic Engineering Club (2020–2021) and coordinated student activities and events.